

DA-VPT: Semantic-Guided Visual Prompt Tuning for Vision Transformers

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Abstract

Visual Prompt Tuning (VPT) has become a promising solution for Parameter-Efficient Fine-Tuning (PEFT) approach for Vision Transformer (ViT) models by partially fine-tuning learnable tokens while keeping most model parameters frozen. Recent research has explored modifying the connection structures of the prompts. However, the fundamental correlation and distribution between the prompts and image tokens remain unexplored. In this paper, we leverage metric learning techniques to investigate how the distribution of prompts affects fine-tuning performance. Specifically, we propose a novel framework, **Distribution Aware Visual Prompt Tuning (DA-VPT)**, to guide the distributions of the prompts by learning the distance metric from their class-related semantic data. Our method demonstrates that the prompts can serve as an effective bridge to share semantic information between image patches and the class token. We extensively evaluated our approach on popular benchmarks in both recognition and segmentation tasks. The results demonstrate that our approach enables more effective and efficient fine-tuning of ViT models by leveraging semantic information to guide the learning of the prompts, leading to improved performance on various downstream vision tasks. The code is released on <https://github.com/Noahsark/DA-VPT>.

1. Introduction

Recent advances in model scaling and dataset expansion [10, 36, 50] have led to powerful vision foundation models, particularly those based on Vision Transformer (ViT) architectures [11]. These models have demonstrated exceptional performance across various computer vision tasks [19, 20, 44]. While fine-tuning these models for downstream tasks like visual recognition [11] or semantic segmentation [27] has become standard practice, the conventional full fine-tuning approach faces significant challenges, including high computational costs, overfitting, and catastrophic forgetting [28, 39]. These challenges have motivated the development

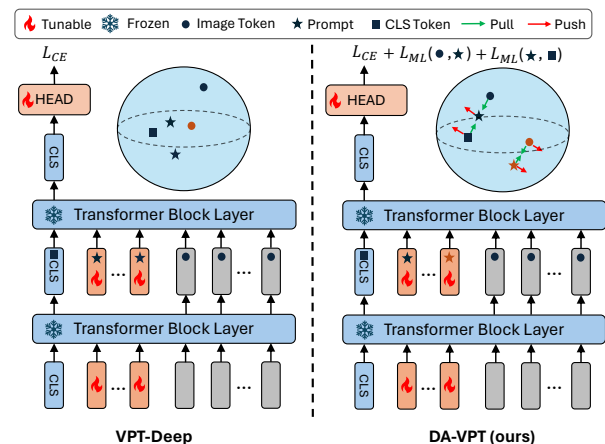


Figure 1. **Comparison between VPT-Deep and DA-VPT.** **Left (VPT-Deep):** Prompts are guided solely by the recognition task, leading to unconstrained distributions between prompts and visual tokens. This allows prompts to attract features from arbitrary classes, potentially hindering the class token’s ability to aggregate class-specific information. **Right (DA-VPT):** Prompts are jointly optimized by the main task and semantic metric learning objectives. The semantic clustering aligns the distributions of prompts, visual tokens, and class tokens, enabling more effective class-specific information aggregation through semantically-guided attention.

of Parameter-Efficient Fine-Tuning (PEFT) methods that selectively update a small subset of model parameters while keeping the majority frozen [3, 23, 25, 28, 32, 42, 47, 63].

Initially emerging from the NLP domain, Houlsby et al. [23] and subsequent works [24, 42] demonstrated that updating a minimal number of parameters could achieve performance comparable to full fine-tuning. These techniques were later adapted to computer vision by Chen et al. [3], who introduced parallel residual networks alongside the ViT backbone. A significant advancement came from Jia et al. [25], who proposed Visual Prompt Tuning (VPT). This method introduces learnable tokens called *visual prompts* at the input level of each ViT layer, effectively aligning downstream task distributions with pre-training data distributions through learnable data-level representations. Building on this foun-

dation, Yoo et al. [62] and Han et al. [17] further enhanced VPT by implementing cross-layer prompt connections with dynamic gating mechanisms, enabling adaptive control of prompt positioning and quantity.

However, existing VPT approaches primarily focus on manipulating prompt connections and structure, while overlooking the intrinsic relationship between prompts and data representations. Specifically, current VPT and its related methods [17, 25, 41, 62] initialize prompts randomly and optimize them solely through downstream task objectives. Although recent work [60] demonstrates improved learning efficiency through data-driven prompt initialization, the potential of leveraging discriminative and class-aware information remains largely unexplored. To deepen our understanding of prompt functionality and distribution, we investigate prompt-token relationships by addressing a fundamental question: **Could prompts be guided to facilitate information flow between image and class tokens to enhance representation learning?**

To address this question, we introduce a novel method that guides VPT optimization by leveraging semantic connections between visual prompts, visual tokens, and class tokens. We propose connecting prompts and visual data by constructing and learning a semantic metric between them in the deep layers of the ViT. For each prompt in these layers, we establish a semantic connection with its closest labeled class. As illustrated in Figure 1, we construct a semantic metric in the latent space by comparing prompts with corresponding image patches. Specifically, we aim to minimize the distance between visual prompts and visual tokens of the same class while maximizing separation from the visual tokens of different classes. We also apply a similar semantic metric between class tokens and prompts.

Our key insight is to increase the likelihood that prompts capture semantic information from same-class visual tokens while filtering out unrelated information. Through semantic metrics in both visual feature and prompt spaces, we demonstrate the effective transfer of relevant semantic information from visual tokens to class tokens via class-specific prompts. In other words, our framework employs related prompts as a *bridge* to connect class tokens and image patch semantic information through guided attention maps.

Extensive experiments across 24 visual recognition tasks in both Fine-Grained Visual Classification (FGVC) [25] and Visual Task Adaptation Benchmark (VTAB-1k) [64] demonstrate substantial improvements over standard VPT. Our method shows consistent effectiveness with both supervised and self-supervised pre-trained models, including MoCo and MAE. Additional evaluations on segmentation tasks further confirm that our approach significantly improves prompt learning efficiency and downstream task performance while requiring fewer prompts and learnable parameters compared to baseline VPT and its related state-of-the-art methods. Our

main contributions are:

- We propose **Distribution Aware Visual Prompt Tuning (DA-VPT)**, a novel framework that enhances prompt learning by constructing semantic metrics between prompts and corresponding image feature patches in deep ViT layers.
- We demonstrate that prompts can effectively bridge semantic information between image patches and class tokens through the attention mechanism, highlighting the importance of guided prompt learning.
- We validate our method’s effectiveness through extensive experiments on 24 visual recognition tasks and 2 segmentation tasks, showing significant improvements over vanilla VPT and its related works for both supervised and self-supervised pre-trained vision models.

2. Related Works

Parameter-Efficient Fine-Tuning (PEFT)

Transformers, initially introduced by Vaswani et al. [55], have revolutionized various domains through pre-training, from natural language processing (e.g., LLaMA [52], GPT [1]) to computer vision (e.g., MAE [21], CLIP [43], ViT-22b [9]). PEFT approaches have emerged to address the computational challenges of fine-tuning these large models by selectively updating only a subset of parameters. Early work by Kornblith et al. [28] focused on training only the classification head, while Zaken et al. [63] demonstrated significant improvements by tuning bias terms alone. Lian et al. [33] and Xie et al. [61] further refined these approaches by introducing adjustable shifting and scaling factors. Another significant direction in PEFT, pioneered by Houlsby et al. [23], involves incorporating lightweight adapter modules alongside Transformer backbones.

Visual Prompt Tuning (VPT) As a prominent branch of PEFT, prompt tuning introduces learnable tokens alongside input data to incorporate task-specific information [31, 32, 34, 35]. Jia et al. [25] pioneered the application of prompts in Vision Transformers (ViT), introducing VPT-Shallow for input layer modification and VPT-Deep for cross-layer integration. This foundational work catalyzed numerous developments in the field: Gao et al. [15] adapted visual prompts for test-time domain adaptation, while Gao et al. [13] extended the approach to video recognition. Subsequent studies enhanced VPT’s capabilities through dynamic mechanisms for optimizing prompt quantity and placement [17, 62], direct connections between intermediate layers and task-specific heads [54], and spatial selection mechanisms for coordinating attention between image patches and visual prompts [41].

Integration of PEFT Approaches While recent comprehensive approaches have demonstrated success in combining multiple PEFT methods [2, 66], our work focuses specifically on integrating bias optimization with VPT, which we empirically found to be sufficiently effective for demonstrat-

ing our method’s capabilities. A comprehensive evaluation of combinations with other PEFT methods lies beyond the scope of this work.

Metric Learning (ML) Metric Learning focuses on learning representations that effectively capture similarities and differences between data samples in the embedding space. Early approaches employed *contrastive loss* to differentiate between class samples [6, 16]. This evolved into *triplet loss* methods that introduce an *anchor* point as a proxy to simultaneously compare positive and negative samples with specified margins [5, 26, 46].

Advanced metric learning techniques have incorporated *Neighbourhood Components Analysis (NCA)* to better understand data distributions and class relationships [26, 38, 48, 49, 51, 57]. Recent studies have demonstrated particular success in applying NCA-based metric learning to Vision Transformer architectures [12, 29, 40], emphasizing the crucial role of data distributions in learning discriminative representations [30, 45, 46, 58]. Further investigations by Tsai et al. [53] and Ren et al. [47] have explored the integration of visual prompts with robust visual perception and deep metric learning.

Building on metric learning, our work examines the interactions between visual prompts, visual tokens, and class tokens within ViTs. We bridge the gap between traditional metric learning techniques and modern visual prompt tuning methods, offering a more principled way to optimize prompt-based transfer learning.

3. Methodology

3.1. Preliminary

The Vision Transformer (ViT) [11] is a fundamental model architecture that applies the original Transformer model [56] to computer vision tasks. Given an input image $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$, ViT divides it into a sequence of N flattened 2D patches, which are then linearly projected into a D -dimensional embedding space. A learnable [CLS] (Class) token $\mathbf{x}_{\text{cls}} \in \mathbb{R}^D$ is prepended to the patch embeddings, serving as a global representation for classification tasks. The resulting sequence of embeddings $\mathbf{X} \in \mathbb{R}^{(N+1) \times D}$ is then passed through L Transformer block layers, where $l \in \{1, \dots, L\}$ denotes the layer index. Each layer consists of a Multi-Head Self-Attention (MHSA) mechanism defined as $\text{MHSA}(\mathbf{X}^l) = \text{Concat}(\mathbf{H}_1, \dots, \mathbf{H}_h)$, where each head $\mathbf{H}_i$ computes a scaled dot-product attention $\text{softmax}(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\mathbf{V})$ with subspaces of Query ($\mathbf{Q}$), Key ($\mathbf{K}$), and Value ($\mathbf{V}$) matrices projected from input embedding $\mathbf{X}^{l-1}$ in the previous layer. The final output is the [CLS] token $\mathbf{x}_{\text{cls}}^L$, used for downstream classification tasks.

Visual Prompt Tuning (VPT) [25] presents a promising PEFT technique for ViT that adapts the pre-trained model to downstream tasks by introducing a small set of

learnable parameters, namely *visual prompts*. In a specific ViT block layer, a sequence of M learnable prompt tokens $\mathbf{P} = \{\mathbf{p}_1, \dots, \mathbf{p}_M\} \in \mathbb{R}^{M \times D}$ is concatenated with the patch embeddings $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\} \in \mathbb{R}^{N \times D}$. Jia et al. [25] propose two VPT settings: **VPT-Shallow** where the prompts are only inserted into the first ViT layer, and **VPT-Deep** where the prompts are appended into every ViT layer. We follow the **VPT-Deep** setting since it has a higher capacity and aligns with our proposed method. The resulting sequence of embeddings $[\mathbf{x}_{\text{cls}}, \mathbf{P}, \mathbf{X}] \in \mathbb{R}^{(M+N+1) \times D}$ is then processed by the next ViT encoder layers. For layer l , the output of the $(l+1)$ -th layer can be described as:

$$[\mathbf{x}_{\text{cls}}^{l+1}, \mathbf{x}_1^{l+1}, \dots, \mathbf{x}_N^{l+1}] = \text{BLK}([\mathbf{x}_{\text{cls}}^l, \mathbf{p}_1^l, \dots, \mathbf{p}_M^l, \mathbf{x}_1^l, \dots, \mathbf{x}_N^l]), \quad (1)$$

where $\mathbf{p}_1^l, \dots, \mathbf{p}_M^l$ are the M prompts in layer l , BLK represents the transformer block, and $[\]$ represents the position reserved for prompts in the next layer. During fine-tuning, only the visual prompts $\mathbf{P}$ and the linear classification head are updated.

Metric Learning (ML) aims to learn a distance metric that captures semantic similarity between data points. The *Neighborhood Component Analysis (NCA)* [49] encourages learned embeddings to have a higher probability of correct classification by nearest neighbor classifiers. Given a set of N labeled data points $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$, where $\mathbf{x}_i \in \mathbb{R}^D$ is the input feature vector and $y_i \in \{1, \dots, C\}$ is the corresponding class label from C classes, the NCA objective is:

$$\mathcal{L}_{\text{NCA}} = - \sum_{i=1}^N \log \frac{\sum_{j \in \mathcal{N}_i} \exp(-D(\mathbf{x}_i, \mathbf{x}_j)/\tau)}{\sum_{k \neq i} \exp(-D(\mathbf{x}_i, \mathbf{x}_k)/\tau)}, \quad (2)$$

where $\mathcal{N}_i = \{j \mid y_j = y_i, j \neq i\}$ denotes the set of neighboring points with the same class, $\tau > 0$ is the temperature parameter, and $D(\cdot, \cdot)$ represents the *cosine similarity*: $D(\mathbf{x}_i, \mathbf{x}_j) = \hat{\mathbf{x}}_i \cdot \hat{\mathbf{x}}_j$ where $\hat{\mathbf{x}} = \frac{\mathbf{x}}{\|\mathbf{x}\|_2}$ represents the L2-normalized vector. Following NCA, recent works [26, 51] introduce learnable class representations $\mathbf{P} = \{\mathbf{p}_i \in \mathbb{R}^D\}_{i=1}^C$, named *proxies*, to represent the C classes. In our work, we propose using prompts in deep layers as proxies for subsets of semantically similar classes.

3.2. Metric Learning on the Learnable Prompts

Our objective is to establish a metric in the feature space that quantifies the distance between learnable prompts and either visual tokens or the [CLS] token. We hypothesize that within each layer, a specific prompt should selectively capture information from a subset of relevant classes rather than searching indiscriminately across the entire class space. This targeted approach enables prompts to become more discriminative in their feature extraction while optimizing the [CLS] token’s ability to aggregate task-specific information from each class effectively. While this structured

To maintain semantic mapping accuracy, we update the mapping after each epoch. During training, we collect and calculate updated class representations $\mathbf{S}$, then update k-means using previous epoch centroids as initialization to adjust the class-prompt mapping.

3.5. Efficient Bias Tuning

To further improve the flexibility in the distribution of visual tokens, we investigate the partial release of ViT backbone bias terms as suggested by Zaken et al. [63]. We found that fine-tuning performance significantly improves when bias terms are partially enabled with our metric guidance loss. The most efficient components are the bias terms $\mathbf{b}_K, \mathbf{b}_V \in \mathbb{R}^D$ in the Key and Value linear projections of the self-attention mechanism (Figure 4b). This observation aligns with findings from Zaken et al. [63] and Cordonnier et al. [7]. Partially allowing bias terms to adapt provides additional flexibility in adjusting visual token distributions and capturing task-specific information under metric guidance.

4. Technical Discussion

4.1. Connection Between Similarity and Attention

In this section, we analyze how changes in token similarity influence attention weights through gradient updates. Specifically, we examine how a small change in the similarity between a prompt $\mathbf{p}$ and a visual token $\mathbf{x}_i$ affects the corresponding attention weight a_i . Let $\Delta\mathbf{p}$ represent a small perturbation that brings $\mathbf{p}$ closer to $\mathbf{x}_i$ in the embedding space. We formalize this relationship in the following theorem:

Theorem 1 (Attention-Similarity Relationship). *For an attention weight perturbation Δa_i computed using the softmax function, the following approximation holds:*

$$\Delta a_i \approx a_i(1 - a_i)\Delta s_i, \quad (6)$$

where Δs_i represents the change in attention score s_i , given by $\Delta s_i = \frac{\Delta\mathbf{p}^\top \mathbf{x}_i}{\sqrt{d}}$, and d is the dimension of the attention head.

This approximation reveals that a positive gradient change in attention weight ($\Delta a_i > 0$) occurs when:

$$a_i(1 - a_i)\Delta s_i = a_i(1 - a_i)\frac{\Delta\mathbf{p}^\top \mathbf{x}_i}{\sqrt{d}} > 0 \quad (7)$$

This condition is satisfied when $\mathbf{p}$ moves closer to $\mathbf{x}_i$ in the embedding space. Conversely, when $\mathbf{p}$ moves away from $\mathbf{x}_i$, Δa_i decreases. This theorem establishes a direct connection between token similarity and attention mapping, demonstrating how our metric learning guidance influences attention through token distribution. The complete proof is provided in the Appendix.

4.2. Analysis of Guided Attention Maps

To further analyze the impact of our metric guidance loss on fine-tuning, we visualize attention maps between visual tokens and prompts across different layers (Figure 3a). Our analysis reveals distinct patterns:

(1) In Shallow Layers:

- Both VPT-Deep and our method show prompts attending to different object subregions
- Our method demonstrates enhanced diversity and precision in information capture

(2) In Deep Layers:

- Attention maps become sparser as token representations become more abstract
- Standard VPT-Deep prompts show limited information selection compared to the [CLS] token
- Our positive prompt ($\ast\mathbf{p}$) successfully identifies informative patches that are subsequently selected by the [CLS] token

These visualizations demonstrate that positively labeled prompts serve as effective "bridges" for semantic information flow to the [CLS] token in deep layers. As shown in Figure 3b, our DA-VPT enables prompts to aggregate discriminative features from same-class data, resulting in more fine-grained attention patterns compared to the baseline. This enhanced information routing significantly improves the model's discriminative capability during fine-tuning.

Artifact Consideration: Recent work by Darcet et al. [8] revealed the existence of attention artifacts in vision transformers, which we also observe in our prompt attention maps (Figure 3a). While they demonstrate that training with *registers* (analogous to learnable prompts) from scratch can eliminate these artifacts, both standard VPT and our method exhibit them due to prompt introduction during fine-tuning rather than pre-training. Nevertheless, our comparative analysis shows that the proposed guidance mechanism reduces both the frequency and spread of artifacts, constraining them more effectively within semantic object boundaries. While our method achieves better alignment between attention patterns and object semantics, the nature and impact of these artifacts on model performance presents an intriguing avenue for future investigation.

4.3. Compatibility with Metric Learning Methods

Our selection of the *Proxy-Anchor* method [26] is motivated by its natural alignment with our hypothesized role of visual prompts, where both proxies and prompts serve as class representatives and comparative anchors for data tokens. Alternative metric learning approaches, such as *Proxy-NCA* [51] and conventional *triplet loss* [5, 22], treat all representations as equal data points. These approaches are less suitable for our framework because the significant disparity between the number of visual prompts (M) and data tokens (N , where $M \ll N$) creates an *unbalanced opti-*

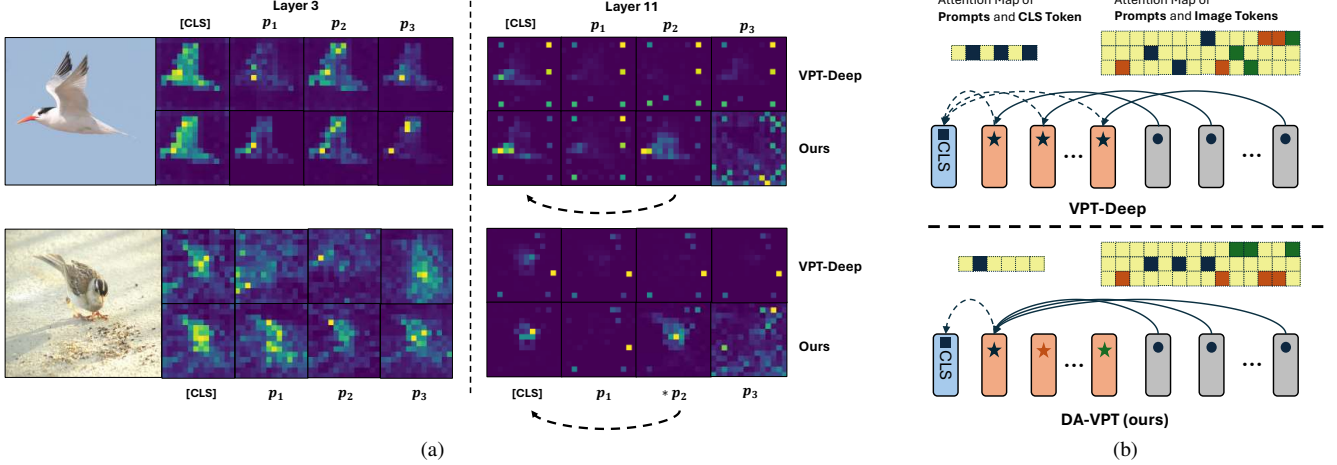


Figure 3. Comparison of attention patterns between VPT-Deep and our method on CUB dataset samples. (a) Attention maps in shallow (layer 3) and deep (layer 11) layers, showing CLS token and sampled prompt attention patterns. In layer 11, $*p$ indicates the prompt guided as positive to the CLS token, while others represent negative prompts. Additional visualization examples are provided in Appendix. (b) Information flow comparison between baseline VPT-Deep and our DA-VPT. Our method enables selected prompts to aggregate and transfer fine-grained details from same-class visual tokens to the [CLS] token more effectively.

mization problem. Our empirical studies further confirm this theoretical intuition: attempting to fine-tune with conventional metric learning methods leads to training instability, whereas the Proxy-Anchor formulation maintains stable optimization by explicitly accounting for the asymmetric nature of prompt-token relationships.

5. Experiments

5.1. Experimental Setup

Datasets. We evaluate our method on three types of visual transfer learning tasks. For visual recognition, we use the Fine-Grained Visual Classification (FGVC) benchmark [25] comprising 5 datasets. For few-shot transfer learning, we employ the Visual Task Adaptation Benchmark (VTAB-1K) [64] containing 19 datasets. Additionally, we evaluate dense prediction tasks on ADE20K [67] and PASCAL Context [37]. Detailed dataset characteristics and experimental settings are provided in the Appendix.

Model Architecture. We employ Vision Transformer (ViT) [11] as our backbone, using the base model **ViT-B** (12 layers) for visual classification tasks and the large model **ViT-L** (24 layers) for semantic segmentation. To evaluate generalization, we initialize the backbone using either supervised pre-training on ImageNet-21K [10] or self-supervised pre-training on ImageNet-1K using methods such as MoCo v3 [4] and MAE [20].

Method Variants. We evaluate two versions of our approach. Our primary method, **DA-VPT**, builds on VPT-Deep [25] while incorporating our proposed metric learning losses $\mathcal{L}_{ML}(\mathbf{X}, \mathbf{P})$ and $\mathcal{L}_{ML}(\mathbf{P}, \mathbf{x}_{cls})$. The enhanced version, **DA-VPT+**, further incorporates efficient bias tuning as detailed in Section 3.5.

Implementation Details. For all experiments, we conduct extensive hyperparameter optimization, including learning rate, parameter decays, and the number of visual prompts for layers both with and without our metric learning guidance. Through this empirical investigation, we identified that the optimal number of prompts for most downstream tasks is approximately 20. For metric learning parameters, we adopt the Proxy-Anchor defaults with margin $\delta = 32$ and temperature $\tau = 10$. Extended experimental details, including hyperparameter studies and ablation analyses, are provided in the Appendix.

5.2. Result Comparison with the State-of-the-Art

We evaluate our method against existing VPT-based and recent related approaches across 24 vision tasks. As shown in Table 1, our DA-VPT+ consistently achieves superior performance over VPT-related methods on both supervised ViT and self-supervised backbones. On ViT-B, DA-VPT+ improves over the VPT-Deep baseline by 2.83 and 4.18 percentage points (pp) on FGVC and VTAB-1K, respectively, and outperforms E2VPT by 2.72 pp and 2.20 pp on these tasks. Notably, even without bias tuning, DA-VPT maintains strong results across major benchmarks. The improvements are particularly pronounced with self-supervised backbones, where our method also surpasses full fine-tuning on all backbones while using fewer parameters. These results highlight the effectiveness and generalizability of our approach across diverse downstream tasks compared to other VPT-based methods.

Table 2 demonstrates our proposed methods, DA-VPT and DA-VPT+, achieve significant improvements over existing baselines and recent competitive methods in semantic segmentation tasks on both the ADE20K and PASCAL Con-

Methods	Mean	FGVC	VTAB-1K			
	Param (M)	Mean Acc (5)	Natural (7)	Specialized (4)	Structured (8)	Mean Acc
ViT-B with Supervised pretrained on ImageNet-21k						
Full	85.98	88.54	75.88	83.36	47.64	68.96
VPT-Shallow	0.11	84.62	76.81	79.68	46.98	67.82
VPT-Deep	0.64	89.11	78.48	82.43	54.98	71.96
E2VPT [17]	0.33	89.22	80.01	84.43	57.39	73.94
DA-VPT (ours)	0.21	91.22	80.25	85.12	58.71	74.69
DA-VPT+ (ours)	0.24	91.94	81.98	86.47	59.96	76.14
ViT-B with MAE pretrained on ImageNet-1K						
Full	85.8	82.80	59.31	79.68	53.82	64.27
VPT-Shallow	0.10	57.84	39.96	69.65	27.50	45.70
VPT-Deep	0.20	72.02	36.02	60.61	26.57	41.73
GateVPT [62]	0.17	73.39	47.61	76.86	36.80	53.09
E2VPT [17]	0.06	—	59.52	77.80	44.65	60.66
DA-VPT (ours)	0.20	82.17	62.14	79.14	54.31	65.19
DA-VPT+ (ours)	0.22	83.20	66.59	82.96	59.28	69.61
ViT-B with MoCo-V3 pretrained on ImageNet-1K						
Full	85.8	84.25	71.95	84.72	51.98	69.55
VPT-Shallow	0.11	79.26	67.34	82.26	37.55	62.38
VPT-Deep	0.20	83.12	70.27	83.04	42.38	65.90
GateVPT [62]	0.17	83.00	74.84	83.38	49.10	69.11
E2VPT [17]	0.11	—	76.47	87.28	54.91	72.88
DA-VPT (ours)	0.21	85.02	74.24	83.21	55.23	70.90
DA-VPT+ (ours)	0.24	86.16	76.86	84.71	58.98	73.53

Table 1. **Comparison of Fine-tuning Methods.** Performance evaluation across 24 vision tasks (5 FGVC and 19 VTAB-1K) using supervised ViT and self-supervised backbones (MAE [20], MoCo-v3 [4]). Detailed per-task results for VTAB-1K are provided in Appendix.

Method	#Param	ADE20K		PASCAL Context	
		mIoU-SS	mIoU-MS	mIoU-SS	mIoU-MS
Full-Tuning	317.3M	47.60	49.18	53.69	55.21
Linear	13.1M	38.09	39.16	46.06	48.13
Bias	13.2M	43.61	45.73	45.15	46.47
VPT (baseline)	13.6M	44.08	46.01	49.51	50.46
SPT-LoRA [18]	14.6M	45.40	47.50	—	—
SPT-Adapter [18]	14.6M	45.20	47.20	—	—
DA-VPT (ours)	13.6M	45.10	47.07	50.15	51.04
DA-VPT+ (ours)	13.7M	46.47	47.21	50.40	51.28

Table 2. **Results of Semantic Segmentation on ADE20K and PASCAL Context.** We report mIoU-SS (single-scale inference) and mIoU-MS (multi-scale inference). All experiments use the ViT-L backbone pre-trained on ImageNet-21K. The #Param column indicates the total number of tunable parameters in the entire framework. For SPT [18], we report the results from the original paper, while for other settings and our baseline, we provide our reproduced results. We highlight the best results other than the full fine-tuning.

text datasets. Compared to classification tasks, dense prediction tasks such as segmentation are much more challenging. Notably, lightweight PEFT methods like Linear or Bias exhibit low efficiency compared to full fine-tuning. In such challenging tasks, our proposed DA-VPT+ still achieves comparable performance while using only 4.3% of the tunable parameters, demonstrating both high parameter efficiency and effectiveness across both datasets.

Table 3 compares state-of-the-art PEFT methods on FGVC [25] using ImageNet-21K pre-trained ViT-B. Our DA-VPT+ achieves the highest mean accuracy of 91.94% across all datasets, surpassing previous SOTA methods SNF [59] and MoSA [65] on FGVC. Notable improvements include gains of 0.6 and 1.8 percentage points on CUB and Cars datasets respectively. Both DA-VPT and DA-VPT+

Method	Dataset	CUB-200 -2011	NABirds	Oxford Flowers	Stanford Dogs	Stanford Cars	Mean Acc (%)	Mean Params (M)
Full fine-tuning [25]		87.3	82.7	98.8	89.4	84.5	88.54	85.98
Linear Probing [25]		85.3	75.9	97.9	86.2	51.3	79.32	0.18
Adapter [23]		87.1	84.3	98.5	89.8	68.6	85.67	0.41
Bias [63]		88.4	84.2	98.8	91.2	79.4	88.41	0.28
AdaptFormer [3]		87.4	84.8	99.0	90.7	81.0	88.58	1.54
VPT-Shallow [25]		86.7	78.8	98.4	90.7	68.7	84.62	0.25
VPT-Deep [25]		88.5	84.2	99.0	90.2	83.6	89.11	0.85
SSF [33]		89.5	85.7	99.6	89.6	89.2	90.72	0.39
SNF [59]		90.2	87.4	99.7	89.5	86.9	90.74	0.25
MP[14]		89.3	84.9	99.6	89.5	83.6	89.38	1.20
E2VPT [17]		89.1	84.6	99.1	90.5	82.8	89.22	0.65
MoSA [65]		89.3	85.7	99.2	91.9	83.4	89.90	1.54
VPT (Baseline)		88.6	85.7	99.2	89.0	87.4	90.14	0.36
DA-VPT (Ours)		90.2	87.4	99.4	89.4	89.7	91.22	0.30
DA-VPT+ (Ours)		90.8	88.3	99.8	89.8	91.0	91.94	0.32

Table 3. **Comparison of various fine-tuning methods on different downstream tasks.** The ViT-B model pre-trained on ImageNet-21K is used as basic backbone. Top-1 accuracy (%) is reported and the best result is in **bold**.

outperform the VPT baseline and full fine-tuning with significant margin while using fewer parameters, demonstrating superior accuracy-efficiency trade-off compared to full fine-tuning and existing PEFT methods.

5.3. Ablation Studies and Discussion

5.3.1. Ablation Study

The ablation study demonstrates the individual and collective contributions of each component in our proposed DA-VPT method on the CUB dataset from the FGVC benchmark and the Natural task category from the VTAB-1k benchmark. The metric learning losses, $\mathcal{L}_{ML}(\mathbf{x}, \mathbf{p})$ and $\mathcal{L}_{ML}(\mathbf{p}, \mathbf{x}_{cls})$, lead to accuracy improvements of 1.08 *pp* on VTAB-1k Natural and 1.22 *pp* on CUB over the baseline. The integration of Efficient Bias further enhances the performance, contributing to an additional 0.97 *pp* and 1.03 *pp* improvement on the respective datasets. When all three components are combined, our DA-VPT method achieves the highest performance, with total accuracy improvements of 2.05 *pp* on VTAB-1k Natural and 2.25 *pp* on CUB.

While the incorporation of these components introduces a minimal increase in latency and memory usage, the gained accuracy far outweighs this slight trade-off. Note that the combination of $\mathcal{L}_{ML}(\mathbf{x}, \mathbf{p})$, $\mathcal{L}_{ML}(\mathbf{p}, \mathbf{x}_{cls})$ and Efficient Bias yields substantial improvements with only a modest increase in parameters. This highlights the efficiency of our method in achieving significant performance gains with minimal parameter overhead.

5.3.2. Analysis of Parameter Impacts

Layer-wise Impact. We first examine the effectiveness of our metric learning loss when applied to different layers. As shown by the blue line in Figure 4a, applying the loss to the final layer yields optimal results in most cases, likely due to the presence of higher-level semantic features in deeper layers. We further investigate the effect of applying our loss across multiple consecutive layers, represented by the

Table 4. **Ablation study on different components in our DA-VPT on two datasets: CUB-200-2011 in FGVC and *Natural* in VTAB-1k.** For each $\mathcal{L}_{ML}$ component, we also search for its optimal hyperparameter. The learnable [CLS] token is combined with Efficient Bias for simplicity. The latency and memory are tested in the same server with RTX4090 GPU.

Components of our Techniques			VTAB-1k <i>Natural</i> (7)		FGVC CUB-200		Latency (ms/img)	Memory (GB)
$\mathcal{L}_{ML}(\mathbf{x}, \mathbf{p})$	$\mathcal{L}_{ML}(\mathbf{p}, \mathbf{x}_{cls})$	Efficient Bias	Param	Accuracy	Param	Accuracy		
				79.45 (base)		88.64 (base)	1.41	2.41
✓			0.14M (0.16%)	79.47 (+0.02) 79.51 (+0.06) 80.53 (+1.08)	0.20M (0.24%)	89.24 (+0.60) 89.06 (+0.42) 89.86 (+1.22)	1.51 1.52 1.54	2.41 2.41 2.41
✓	✓	✓	0.16M (0.19%)	80.06 (+0.61) 81.02 (+1.57) 81.50 (+2.05) 81.98 (+2.53)	0.23M (0.27%)	89.55 (+0.91) 90.41 (+1.77) 90.54 (+1.90) 90.89 (+2.25)	1.45 1.53 1.53 1.56	2.76 2.76 2.76 2.76

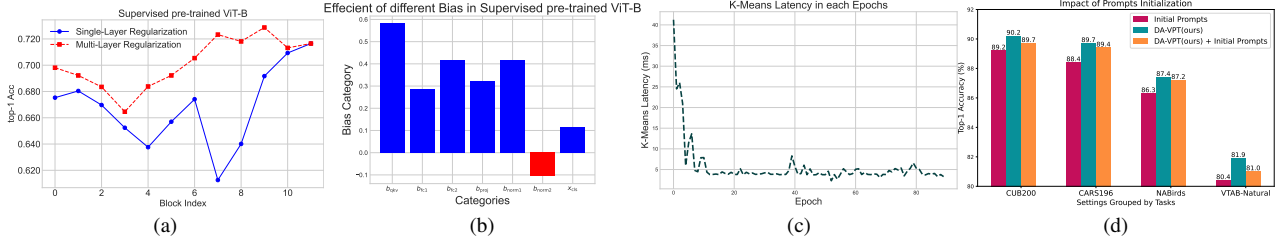


Figure 4. **4a** Illustrates the impact of the number and position of the layers to which the proposed metric learning loss is applied. **4c** This figure shows the latency of the k-means calculation in each epoch. **4b** Illustrates the importance of each category of efficient bias measured on the CUB-200-2011 dataset. **4d** This figure shows the comparison of the performance with or without the prompts initialization with data mean value.

red line, which shows the performance when applying the loss from a specific layer through to the final layer. The impact varies notably across different pre-trained models, with detailed results for MAE and MoCo provided in the Appendix.

Efficient Bias Components. Figure 4b demonstrates that specific categories of efficient bias contribute disproportionately to performance improvements. This observation underscores the importance of selective optimization of bias components rather than uniform adjustment across all parameters.

Semantic Mapping Updates. The computational cost of k-means clustering for semantic mapping updates is illustrated in Figure 4c. Notably, while the initial epochs incur higher computational overhead, the latency decreases significantly in later epochs as class representations stabilize. This suggests that the computational cost of maintaining dynamic class-prompt mappings becomes negligible as training progresses.

Prompt Initialization. We investigate the impact of prompt initialization by comparing mean value initialization on both baseline VPT and our proposed DA-VPT, following the methodology of Wang et al. [60], where prompts are initialized with *mean pooling* values from the dataset at each layer. As shown in Figure 4d, our analysis reveals that such initialization actually impedes our method’s effectiveness. We attribute this to the increased difficulty in

guiding prompts to capture discriminative information when initialized with homogeneous content from the mean value. Additional parameter impact analyses are provided in the Appendix.

6. Conclusion

This paper introduces Distribution-Aware Visual Prompt Tuning (DA-VPT), a novel framework that improves visual prompt learning in Vision Transformers (ViT) through semantic metric construction between prompts and image features. Our method guides prompts to serve as effective bridges for semantic information flow between image patches and class tokens via the attention mechanism. Extensive evaluations across 24 visual recognition and 2 segmentation tasks demonstrate that DA-VPT significantly outperforms vanilla VPT and other related methods while using fewer prompts and parameters. Our results highlight the importance of considering the intrinsic connection between visual prompts and data samples and showcase the potential of our approach to enhance the transfer learning capabilities of pre-trained vision models. We believe that our findings can inspire further research on parameter-efficient fine-tuning strategies and contribute to the development of more effective and efficient vision foundation models.

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